# NAIL MAMMADZADA

(520) 248-5116 | n.mamzada@gmail.com | linkedin.com/in/nailmammadzada | github.com/nmamzada

## EDUCATION

**University of Arizona** **Expected May 2028**

*B.S. Mathematics | B.S. Statistics & Data Science | Minor in Business Administration | GPA: 3.8*

- **Coursework:** Linear Algebra, Vector Calculus, Statistics & ML , Quantitative Financial Management, Theory of Probability, Theory of Statistics, Stochastic Processes, Real Analysis, Partial Differential Equations
- **Honors:** 1st Place, National Mathematics Olympiad (2024); 1st Place (Team), World Scholar's Cup (2023)

## EXPERIENCE

**SOCAR Upstream Management International** **Baku, Azerbaijan**

*Machine Learning Intern, Drilling Department & Asset Management* *Summer 2026*

- Built an NLP classifier that sorts drilling logs into 22 activity types with 95–98% accuracy vs. engineer labels
- Wrote a PDF-to-SQL pipeline (pdfplumber, SQLite) that replaced manual entry of daily drilling reports
- Built a Claude API tool that writes SQL from plain-English questions, cutting data requests from 2 hours to 2 minutes

*Data Science Intern, Subsurface Department* *Summer 2025*

- Built machine learning models to forecast oil production across 150+ active wells
- Developed an Arps decline-curve tool in SciPy that improved long-term forecast and economic break-even accuracy 40% in backtests

## LEADERSHIP

**Quantitative Finance Club, University of Arizona** | Founder & President **2024 – Present**

- Founded and lead a 40+ member club with four teams: quant development, research, trading, and risk
- Teach weekly sessions on Python, statistics, valuation, options pricing and backtesting
- Organized 4+ industry events with professionals from JPMorgan Chase, Intel and Amazon

## PROJECTS

**Options Pricing & Valuation Dashboard** | *Python, Dash, scikit-learn* *2025*

- Priced European options with Black–Scholes and valued stocks with a DCF model on live market data
- Simulated GBM price paths and tested random-forest return forecasts against a random-walk baseline

**LSTM: Walk-Forward Equity Algorithm** | *Python, TensorFlow/Keras, Alpaca API, Pandas* *2026*

- Built an LSTM forecasting 5-day stock returns, sizing a long-only book by inverse-volatility targeting, validated in a walk-forward backtest with transaction costs.
- Deployed it as an automated Alpaca paper-trading bot with one research/live pipeline, model persistence, and unit-tested features

## SKILLS & INTERESTS

**Programming:** Python (pandas, NumPy, SciPy, scikit-learn), SQL, R, TypeScript, Git,

**Quant & Finance:** Monte Carlo simulation, Black–Scholes, DCF valuation, factor models, portfolio optimization, back testing, Algorithmic Trading, Machine Learning

**Interests:** Poker, chess, golf, reading